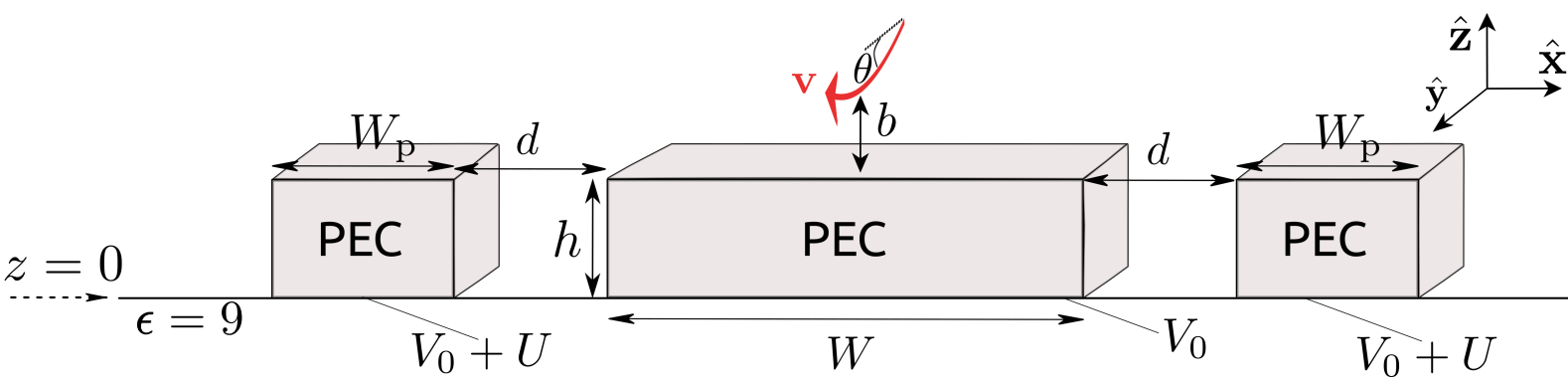


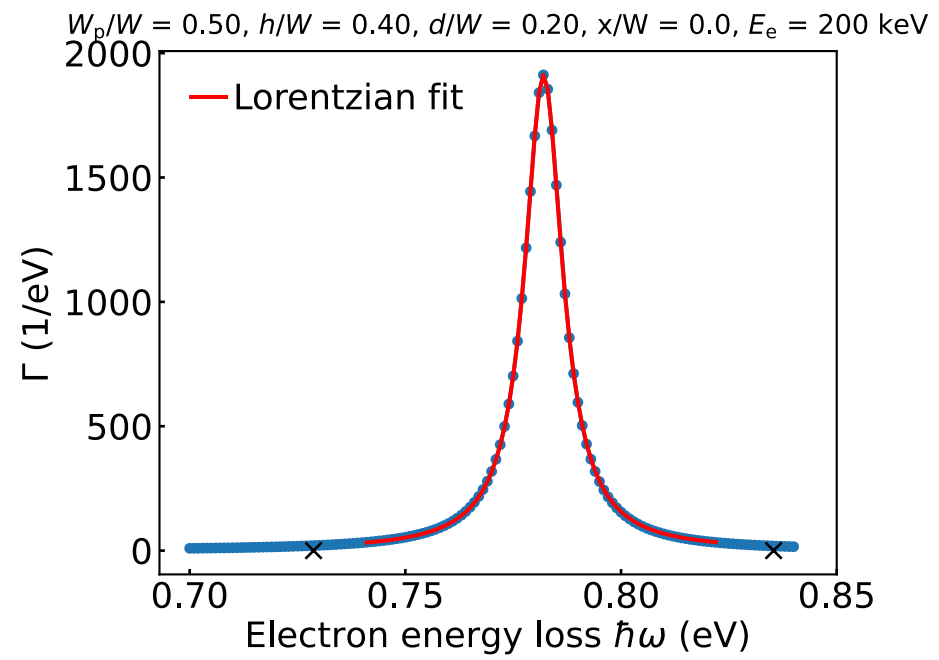
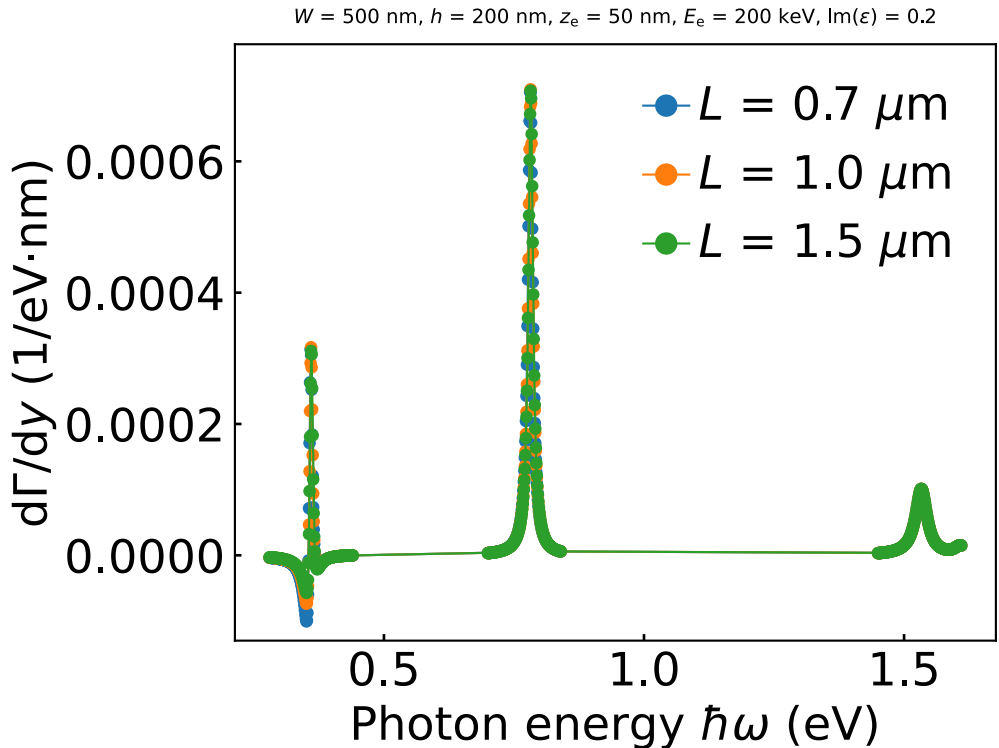
Electron-coupling to waveguide

$$\Gamma_{\text{EELS}}(\omega) = 2 \int_{b_{\text{min}}+h}^{\infty} \frac{vdz}{\sqrt{\left| c^2 \beta^2 \sin^2 \theta + \frac{2eV(x,z)}{m_e \gamma_e} \right|}} \frac{d\Gamma_{\text{EELS}}(\omega, x, z)}{dy} \Big|_{z=0}$$

from BEM2D



$E_e = 200 \text{ keV}$
 $W_p/W = 0.5$
 $d/W = 0.2$
 $x/W = 0$
 $h/W = 0.4$
 $W = 500 \text{ nm}$
 $\epsilon_{\text{Si}} = 12.25 + i0.2$



$$0 = \frac{V(z_{\text{min}})}{U} \frac{U}{V_0} + \frac{m_e c^2 \gamma_e}{2e} \frac{\beta^2 \sin^2 \theta}{V_0} \rightarrow$$

